

CPSC 250

More on Fibonacci and Complex Numbers

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1 Motivation

The Fibonacci sequence is surprisingly rich.

Previously, we introduced the sequence and discussed several algorithms for computing it.

In this lecture, we return to Fibonacci from a more mathematical point of view.

This gives us a chance to discuss:

- alternate indexing conventions,
- negative Fibonacci numbers,
- Binet's formula,
- non-integer inputs,
- complex numbers in Python,
- and plotting mathematically interesting functions.

2 The Common Fibonacci Convention

A common version of the Fibonacci sequence begins:

$$1, 1, 2, 3, 5, 8, 13, \dots$$

with:

$$f_1 = 1$$

$$f_2 = 1$$

and:

$$f_n = f_{n-1} + f_{n-2}$$

for:

$$n \geq 3$$

This convention uses natural-number indexing.

3 The Zero-Indexed Convention

Earlier, we used a slightly different convention:

$$0, 1, 1, 2, 3, 5, 8, \dots$$

with:

$$f_0 = 0$$

$$f_1 = 1$$

and:

$$f_n = f_{n-1} + f_{n-2}$$

for:

$$n \geq 2$$

This convention is common in programming because indexing often begins at zero.

4 Extending Fibonacci Backward

A natural question is:

Can we extend the Fibonacci sequence to negative indices?

The answer is yes.

Using the recurrence:

$$f_n = f_{n-1} + f_{n-2}$$

we can solve backward.

Since:

$$f_1 = f_0 + f_{-1}$$

and:

$$1 = 0 + f_{-1}$$

we obtain:

$$f_{-1} = 1$$

Next:

$$f_0 = f_{-1} + f_{-2}$$

so:

$$0 = 1 + f_{-2}$$

and therefore:

$$f_{-2} = -1$$

Continuing gives:

$$\dots, 89, -55, 34, -21, 13, -8, 5, -3, 2, -1, 1, 0, 1, 1, 2, 3, 5, \dots$$

The signs alternate for negative indices.

5 Ratio Behavior

For positive indices, the ratio:

$$\frac{f_{n+1}}{f_n}$$

approaches:

$$\phi = \frac{1 + \sqrt{5}}{2}$$

as:

$$n \rightarrow +\infty$$

For negative indices, a related limiting behavior occurs:

$$\frac{f_{n+1}}{f_n} \rightarrow -\frac{1}{\phi}$$

as:

$$n \rightarrow -\infty$$

Thus, the golden ratio appears on the positive side, while its negative reciprocal appears on the negative side.

6 Binet's Formula

Binet's formula is:

$$f_n = \frac{\phi^n - \left(-\frac{1}{\phi}\right)^n}{\sqrt{5}}$$

This formula works beautifully for integer values of n , including negative integers.

7 A Strange Question

Once we have a formula, we can ask a strange but interesting question:

What happens if n is not an integer?

For example:

$$f_{0.5}$$

Using Binet's formula:

$$f_{0.5} = \frac{\phi^{0.5} - \left(-\frac{1}{\phi}\right)^{0.5}}{\sqrt{5}}$$

But:

$$a^{0.5} = \sqrt{a}$$

so this requires:

$$\sqrt{-\frac{1}{\phi}}$$

which is the square root of a negative number.

This leads naturally to complex numbers.

8 Complex Numbers

The imaginary unit is defined by:

$$i = \sqrt{-1}$$

A complex number has the form:

$$z = a + ib$$

where:

- a is the real part,
- b is the imaginary part.

The magnitude of a complex number is:

$$|z| = \sqrt{a^2 + b^2}$$

9 Complex Numbers in Python

Python uses:

j

instead of i for the imaginary unit.

This is common in engineering.

Example:

```
import cmath

a = 4
b = 3

z = complex(a, b)

print(f"The real part of z is {z.real}")
print(f"The imaginary part of z is {z.imag}")
```

This creates:

$$z = 4 + 3j$$

10 Polar Form

The module `cmath` can convert a complex number to polar form.

```
w = cmath.polar(z)

print(f"The size and phase of z are {w}")
```

The result contains:

- magnitude,
- phase angle.

The phase angle is given in radians.

11 Converting Back to Rectangular Form

We can also convert from polar form back to rectangular form:

```
z3 = cmath.rect(w[0], w[1])

print(z3)
```

12 Plotting Fibonacci for Real Values

Now we are ready to ask Python to evaluate Binet's formula for many real values of n , not just integers.

The goal is to make a plot of:

$$f_n$$

versus:

$$n$$

for real values of n between, for example:

$$-30 \leq n \leq 30$$

13 Important Practical Note

For non-integer values of n , Binet's formula may produce complex values.

Therefore, we may need to plot:

- the real part,
- the imaginary part,
- or both.

14 Plotting Plan

A reasonable plan is:

1. Define the Fibonacci function using Binet's formula.
2. Generate a list or array of n -values.
3. Compute the corresponding Fibonacci values.
4. Plot the normal integer Fibonacci values as distinct points.
5. Plot the real and imaginary parts for real-valued input.
6. Draw the horizontal and vertical axes.

15 Matplotlib Axis Lines

Matplotlib provides:

```
plt.axvline(x=0)
plt.axhline(y=0)
```

These draw:

- a vertical line at $x = 0$,
- a horizontal line at $y = 0$.

16 Key Takeaways

- Fibonacci numbers can be indexed in more than one way.
- The sequence can be extended to negative indices.
- Negative-index Fibonacci numbers alternate signs.
- Binet's formula works for integer indices.
- Extending the formula to non-integer inputs naturally introduces complex numbers.
- Python supports complex numbers directly.
- The `cmath` module provides tools for complex arithmetic.